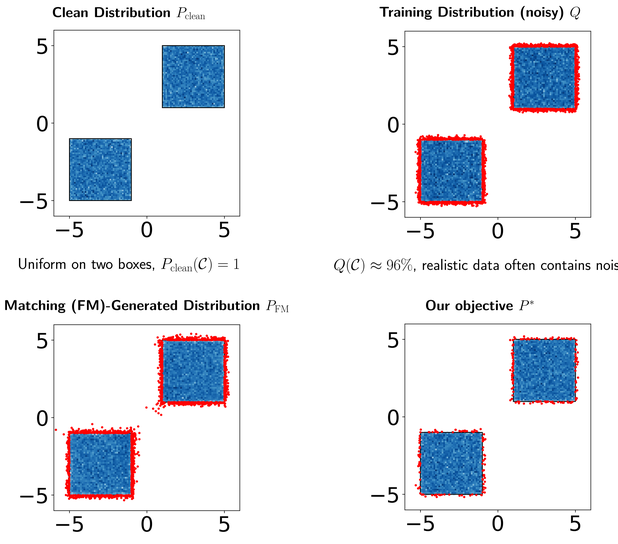
Primal-Dual Flow Matching for Sample-Wise Constrained Generation

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Constraint-Aware Generation

- **Motivation:** Consider the problem of generating samples that satisfy certain constraints $\mathcal{C}$, a generative model trained on either clean or noisy samples can produce infeasible samples $x_{\text{gen}} \notin \mathcal{C}$.



Samples violating the constraint $\mathcal{C}$ are denoted in red. The target satisfaction rate is $b = 99.5\%$.

- **Limitations of existing works:** (1) Requirements on $\mathcal{C}$, e.g., convexity and distance function. (2) Implicit control over constraint satisfaction rate without a user-defined target. (3) Lack of theoretical guarantees.

Problem formulation

Let $x \in \mathbb{R}^d$ denote a sample. Q is the training distribution and P is the modeled distribution with density p. $\mathcal{C} \subset \mathbb{R}^d$ denotes the constraint set and $b \in (0, 1)$ denotes the target satisfaction rate. The population-level problem is given as

$$\underset{P}{\text{minimize}} \quad \mathbb{E}_{x\sim Q}[-\log p(x)] \text{s.t.} \quad b - P(\mathcal{C}) \le 0, \qquad \text{(OPT)}$$

in which $P(\mathcal{C}) = \int_{\mathcal{C}} p(x)\mathrm{d}x$ and the $\mathcal{C}$ is ONLY accessible via membership oracle,

$$\mathbf{1}_{\mathcal{C}}(x) = \begin{cases} 1, & x \in \mathcal{C}, \\ 0, & x \notin \mathcal{C}. \end{cases}$$

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Primal-Dual Flow Matching (PDFM)

Define the Lagrangian

$$L(p, \lambda) = \mathbb{E}_{x\sim Q}[-\log p(x)] + \lambda \left(b - \int_{\mathcal{C}} p(x)\,\mathrm{d}x \right),$$

in which q: the density of Q; $\mathcal{C}^c$: the complement of $\mathcal{C}$; λ: the dual variable.

Replacing $\mathbb{E}_{x\sim Q}[-\log p(x)]$ by the FM loss leads to the practical Lagrangian:

$L'(u_\theta, \lambda') = \mathcal{L}_{\text{FM}}(\theta) + \lambda'\big(b - P_{u_\theta}(\mathcal{C})\big), \quad \mathcal{L}_{\text{FM}}(\theta) = \mathbb{E}_{t,x_0,x_1}\big[\|u_\theta(x_t, t) - (x_1 - x_0)\|^2\big],$

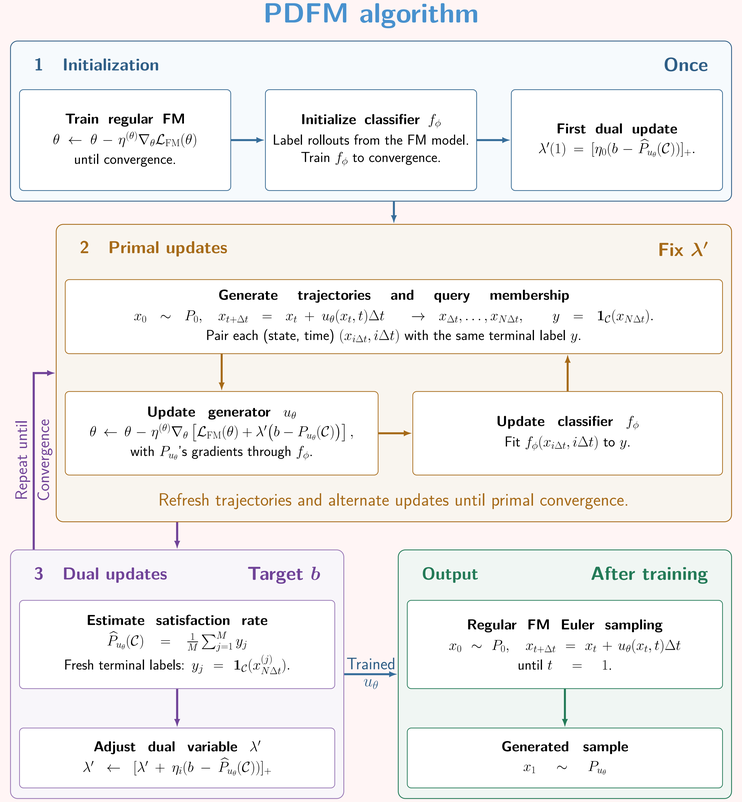
in which u_θ: the parameterized velocity field; P_{u_θ}: u_θ's generated distribution.

(Proposition 4.1) Computation of $\nabla_\theta P_{u_\theta}(\mathcal{C})$.

$$\nabla_\theta P_{u_\theta}(\mathcal{C}) \approx N\mathbb{E}_{i,x_{i\Delta t}}\left[\nabla_\theta u_\theta(x_{i\Delta t}, i\Delta t)\nabla_u f_\phi(x_{i\Delta t} + u\Delta t, (i+1)\Delta t)|_{u=u_\theta(x_{i\Delta t}, i\Delta t)}\right].$$

- $\Delta t = 1/N$ for N Euler steps, $i \sim \text{Unif}\{0, \ldots, N-1\}$.
- $x_{i\Delta t}$ is the Euler rollout up to time $i\Delta t$ following u_θ.
- f_ϕ is a trained classifier that predicts from the x_t and t whether following u_θ will end in $\mathcal{C}$ at $t = 1$.

PDFM algorithm

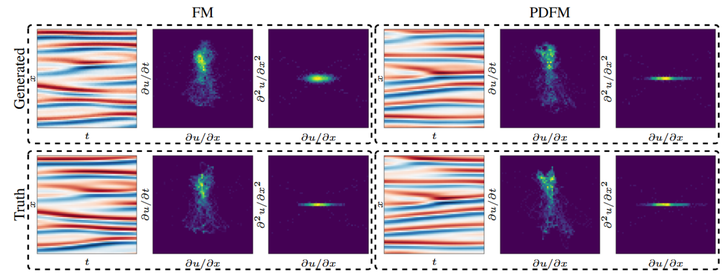


PDE-Constrained Generation

Objective: Generate chaotic dynamics with both physical consistency and statistical fidelity.
Setting: Generate 64×64 Kuramoto–Sivashinsky (KS) space–time samples.
Feasibility: $\delta_{L_1}(x) < \tau$. δ_{L_1} is the mean absolute one-step solver residual (membership oracle).
Case 1: Clean data, $Q(\mathcal{C}) = 1$; $\tau = 0.033$, target $b = 98\%$.
Case 2: Spatially degraded data (linear interpolation), $Q(\mathcal{C}) = 39.47\%$; $\tau = 0.05$, target $b = 90\%$.

	Case 1 ($Q(\mathcal{C}) = 1$)			Case 2 ($Q(\mathcal{C}) \approx 39.47\%$)			
Method	$P(\mathcal{C})$ (%) ↑	δ_{L_1} ↓	HE ↓	$P(\mathcal{C})$ (%) ↑	δ_{L_1} ↓	HE_{clean} ↓	HE_{deg} ↓
FM(Q)	14.19	0.0409	(0.0087, 0.0453)	9.82	0.0622	(0.0094, 0.0465)	(0.0086, 0.0565)
FM($Q \mid \mathcal{C}$)	—	—	—	33.24	0.0539	(0.0135, 0.0477)	(0.0114, 0.0489)
PCFM	6.03	0.0484	(0.0156, 0.1547)	0.06	0.1022	(0.0173, 0.0729)	(0.0186, 0.0823)
FM-RE	43.17	0.0339	(0.0117, 0.0329)	46.33	0.0513	(0.0118, 0.0558)	(0.0097, 0.0570)
PDFM	**97.74**	**0.0235**	(0.0107, 0.0269)	**90.48**	**0.0406**	(0.0118, 0.0435)	(0.0103, 0.0457)

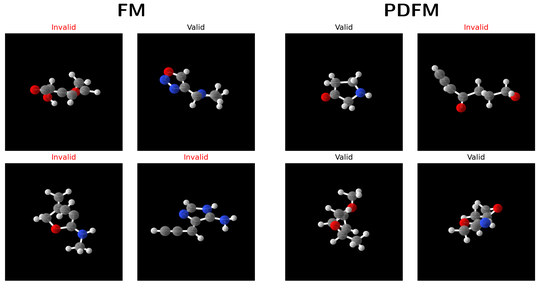
Means over 10 trials, with 2,000 generated samples per method per trial.
HE: mean absolute difference between generated/reference histograms. Pairs: $(\partial_x u, \partial_t u)$ and $(\partial_x u, \partial_{xx} u)$.



Case 1. Generated samples and derivative histograms (top); corresponding solver rollouts (bottom).

Molecule Generation

Objective: Generate stable 3-D molecules.
Setting: Training on the QM9 dataset, generate new molecules, in which every atom is stable.
Feasibility: Valid atomic valencies after bond inference (membership oracle).



Method	Stable (%) ↑	NLL ↓
FM	32.77	−19.74
PDFM	**48.75**	**−20.70**

Means over 10 trials. Per trial: 1,000 generated molecules for stability; 100 randomly selected training samples for NLL. NLL measures likelihood fit (lower is better). PDFM improves stability and NLL.

Conclusion: PDFM

- combines provable feasibility guarantees with practical primal-dual training using membership feedback;
- achieves competitive constraint satisfaction and distribution fidelity across tasks.

https://github.com/ZhengyanHuan/PDFM